

Aritra Konar

Burdwan, West Bengal, India

+91 98325 05937 | konararitra72@gmail.com | github.com/Aritra203 | linkedin.com/in/aritra-konar03 |  Aritra Konar

Experience

Springer Capital, SDE Intern | Full-Time

March 2026 - present

- **Technologies**- Software Development, Front-End Development

Coding Junction President | Part-Time

May 2025 - present

- Designed **Websites**, Guiding the students and mentoring them.
- Developed and deployed **Websites and Portals** integrating **New-age Frameworks** to solve real-world problems.
- **Technologies**- HTML5, CSS3, Javascript, ReactJS, Git

NORDEK, Campus Ambassador | Part-Time

Jan 2024 - Sep 2024

- **Technologies**- Web3, Solidity, Smart Contracts

Education

8.30/10.0 **B.E. in Information Technology**, University Institute of Technology | The University of Burdwan

2022-26

85.8% **Higher Secondary(12th Boards)**, Burdwan CMS High School | Burdwan, India

2021-22

95.28% **Secondary(10th Boards)**, Burdwan CMS High School | Burdwan, India

2019-20

Achievements: Winner at ClashDevs events in TECHFIESTA 2024 | Hacker at Hack4Bengal 2.0 | Contributor at Hacktoberfest 2023 | Top KubeCaptian at Cloud Native Hooghly | Top Campus Ambassador for GSSOC 2024 | Trainee at IIRS, ISRO

Honors: Postman API Fundamentals Student Expert URL: [Click Here](#)
| Postman Student Leader URL: [Click Here](#)

Skills

Languages	C, Java, Python, SQL, Git, HTML, JavaScript
Frameworks	CSS, Bootstrap, MaterialUI, Tailwind, Firebase, Github, Bubble, Framer, NextJs
Soft Skills	Time Management, Problem-solving, Documentation, Engaging Presentation, Leadership, On-site coordination & Teamwork.

Projects

Security Enhancement of AES Algorithm Using Dynamic Key Generation and Dynamic Internal Transformation

Aug 2025 - Present

The Enhanced AES-128 Encryption System is a research-grade implementation that eliminates static behavior in traditional AES by introducing multi-layered dynamic transformations.

- Designed and implemented an advanced **AES-128** encryption system with multi-layered dynamic transformations to eliminate static cryptographic patterns.
- Developed **session-based key generation** using environmental entropy, time-based randomness, and iterative **SHA-256** key stretching.
- Built dynamic S-Box generation using **chaotic maps** (logistic & tent) to improve resistance against linear and differential **cryptanalysis**.
- Implemented state-dependent **ShiftRows** and key-dependent **MixColumns**, ensuring invertibility and enhanced diffusion.
- Achieved ~50% **avalanche effect** and improved brute-force complexity ($\sim 2^{140}$ effective security).

MONAD Chain Game: The Future of Blockchain Gaming

Apr 2025 - May 2025

Unleash the power of MONAD's revolutionary blockchain technology in an immersive card battle experience with parallel execution, state synchronization, and RaptorCast NFT propagation.

- Architecture and Technology: The game is built with a **React** and **TypeScript** frontend and a **Node.js**, **WebSocket** backend for real-time interaction. It deeply integrates with the MONAD blockchain using **Ethers.js** and **Hardhat**, specifically leveraging MONAD's unique features like the **Parallel Execution Engine**, **StateSync** for efficient data transfer, and the **RaptorCast** protocol for reliable NFT propagation.
- Key Achievements and Metrics: By harnessing MONAD's technology, the game achieves significant performance milestones. It delivers 50-100x faster gameplay with sub-second finality, reduces data transfer by 95% to make the game playable on 2G networks, and cuts transaction costs by 70% through batched operations. The system is robust enough to maintain 99.9% reliability even when 40% of network nodes are offline.

Arogya X - Comprehensive Hospital Management System

Jun 2025 - Jul 2025

Arogya X is a state-of-the-art Hospital Management System designed to revolutionize healthcare operations.

- The system is built with a modern tech stack, featuring **React** with **Vite** and **Tailwind CSS** on the frontend, and a robust **Node.js** backend with **MongoDB** for the database. Security is handled through **JWT authentication** and **Bcrypt** for password hashing.

Languages

English	Professional proficiency
Bengali	Native proficiency
Hindi	Basic proficiency